

MARSABIT, Kenya

WMO: 636410

Lat: 2.300N

Lon: 37.900E

Elev: 4413

StdP: 12.50

Time Zone: 3.00 (E03)

Period: 90-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	57.6	58.7	40.3	43.3	73.4	44.9	51.8	73.6	55.8	62.7	47.6	62.9	26.5	160	0.917

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.0	82.1	63.9	80.7	63.6	79.5	63.6	68.4	74.8	67.6	73.9	67.1	73.4	15.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	115.1	70.4	65.9	112.8	69.9	65.2	110.1	69.4	35.6	75.0	34.9	74.2	34.4	74.0	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.3	43.0	39.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.5	71.2	72.5	72.1	70.6	70.1	68.0	66.3	66.6	68.4	69.7	69.0	69.3
	DBStd	2.88	2.60	2.48	2.41	2.36	1.64	1.74	2.05	2.13	1.99	2.34	2.05	2.06
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	24	1	0	0	0	0	1	10	7	1	1	2	1
	CDD50	7110	657	631	686	619	624	540	504	516	552	612	569	600
	CDD65	1660	194	211	221	169	159	91	50	58	103	148	121	135
	CDH74	3991	530	895	869	412	247	113	48	90	215	336	99	137
	CDH80	191	12	71	59	23	4	2	2	2	5	9	0	2
Wind	WSAvg	22.6	19.6	22.6	22.1	21.1	23.2	23.2	24.3	24.6	25.1	23.8	19.9	21.6
Precipitation	PrecAvg	26.9	1.3	1.1	1.8	7.0	2.8	0.4	0.3	0.4	0.1	2.9	4.4	2.8
	PrecMax	49.0	10.9	9.1	4.8	15.0	9.6	2.7	1.1	1.6	0.5	13.0	16.3	12.5
	PrecMin	5.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0
	PrecStd	11.1	2.5	2.2	1.6	4.4	2.9	0.6	0.3	0.4	0.1	3.2	3.5	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.6	84.1	83.6	82.6	79.9	79.2	77.1	78.9	80.2	80.8	78.7	80.0
		MCWB	61.9	62.7	64.1	64.7	65.5	64.1	62.6	63.4	63.2	64.2	64.6	62.5
	2%	DB	80.0	81.8	81.8	80.2	77.9	76.9	75.3	76.0	77.7	78.8	76.7	77.7
		MCWB	62.5	63.0	64.4	65.4	65.1	63.2	61.9	61.7	62.1	63.7	65.3	63.1
	5%	DB	78.5	80.5	80.3	78.3	76.6	75.3	73.9	74.6	76.4	77.3	75.2	75.9
		MCWB	62.4	62.8	64.1	65.7	65.3	62.5	61.4	61.1	61.6	63.4	65.3	63.3
	10%	DB	77.2	79.1	78.7	76.4	75.3	73.9	72.7	73.3	75.1	76.0	73.8	74.3
		MCWB	62.2	62.7	64.1	65.7	65.1	62.1	61.0	60.7	61.2	63.0	65.1	63.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.5	68.1	69.0	69.2	69.1	65.9	65.8	64.9	65.1	67.4	68.4	67.8
		MCDB	74.6	76.9	76.4	75.3	75.6	74.9	74.1	74.8	76.4	73.2	73.8	73.2
	2%	WB	66.5	66.6	67.6	68.3	67.8	64.8	63.7	63.5	63.8	66.3	67.3	67.1
		MCDB	73.9	75.8	75.0	74.3	74.3	74.0	72.6	73.1	74.3	71.9	72.7	72.2
	5%	WB	65.7	65.7	66.9	67.5	66.9	63.9	62.6	62.3	62.9	65.6	66.6	66.4
		MCDB	73.2	75.1	74.4	73.4	73.4	72.8	71.2	71.9	73.3	71.3	71.8	71.4
	10%	WB	65.0	64.9	66.2	67.0	66.2	63.1	61.7	61.5	62.1	64.9	66.0	65.7
		MCDB	72.5	74.4	73.9	72.7	72.6	71.5	70.3	71.0	72.2	70.9	71.0	70.5
Mean Daily Temperature Range	5% DB	MDBR	12.1	14.0	13.6	11.5	12.0	13.8	14.1	15.0	15.7	13.2	10.0	9.7
		MCDBR	13.6	15.4	15.8	14.6	13.2	15.0	15.0	16.3	16.6	15.3	11.9	12.4
		MCWBR	6.5	5.8	5.2	5.0	4.6	5.1	5.1	5.9	5.2	4.8	4.4	5.0
	5% WB	MCDBR	11.8	13.8	12.8	11.1	11.5	14.3	14.2	15.5	16.0	12.1	9.9	9.4
		MCWBR	5.7	5.5	5.4	5.1	5.0	5.7	5.8	6.1	5.8	5.0	4.8	4.8
Clear-Sky Solar Irradiance	taub	0.342	0.362	0.372	0.358	0.355	0.412	0.415	0.375	0.353	0.347	0.356	0.349	
	taud	2.515	2.447	2.437	2.525	2.533	2.307	2.287	2.411	2.479	2.539	2.514	2.516	
	Ebn at Noon	312	306	301	299	293	272	273	290	303	307	305	308	
	Edh at Noon	35	38	38	34	33	40	42	38	36	34	35	34	
All-Sky Solar Radiation	RadAvg	2055	2137	2063	1862	1950	1857	1830	1964	2114	1933	1743	1864	
	RadStd	103	74	132	86	53	75	78	60	54	75	76	119	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page